Enhancing Self-Supervised Video Representations with HAF-STD Net: A Comparative Evaluation Against VLM for Action Recognition



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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
BB	Balance Beam
CNN	Convolutional Neural Network
FX	Floor Exercise
GPU	Graphics Processing Unit
HMDB	Human Motion DataBase
JEPA	Joint Embedding Predictive Architecture
KNN	K-Nearest Neighbors
MAE	Masked Autoencoder
MCC	Matthews Correlation Coefficient
MIT	Master of Information Technology
MLP	Multi-Layer Perceptron
NF4	Normal Float 4-bit (Quantization Format)
PCA	Principal Component Analysis
RBF	Radial Basis Function
SDPA	Scaled Dot-Product Attention
SSL	Self-Supervised Learning
SVM	Support Vector Machine
UB	Uneven Bars
V-JEPA	Video Joint Embedding Predictive Architecture
ViT	Vision Transformer
VLM	Vision-Language Model
VT	Vault

Chapter 1

Introduction

1.1 Introduction

Video understanding represents one of the most challenging frontiers in computer vision and artificial intelligence research. Unlike static image recognition, which has seen remarkable progress over the past decade, video comprehension demands that models capture not only spatial features within individual frames but also the complex temporal dynamics that unfold across sequences of frames (Bardes et al., 2024). This temporal dimension introduces an exponential increase in computational complexity and representational challenges, making video understanding a substantially more demanding task than its image-based counterpart.

Action recognition—the task of identifying human activities from video data—is a central problem in video understanding with broad applications spanning sports analytics, surveillance, human-computer interaction, healthcare monitoring, and content-based video retrieval (Bordes et al., 2024). The difficulty of action recognition varies dramatically depending on the granularity of the target classes: coarse-grained recognition of clearly distinct activities (e.g., "running" vs. "swimming") differs fundamentally from fine-grained recognition of visually similar activities (e.g., distinguishing between gymnastics events that occur in similar arenas).

Two fundamentally different paradigms have emerged for learning video representations in recent years. Self-Supervised Learning (SSL) enables models to learn representational features from unlabeled video data by solving predictive latent pretext tasks, such as predicting masked spatiotemporal regions without pixel reconstruction (Assran et al., 2023, 2025). Vision-Language Modeling (VLM), on the other hand, pre-trains models on massive multimodal paired visual and textual datasets using visual instruction tuning and autoregressive language modeling objectives (Bordes et al., 2024; Qwen Team, 2025).

Both paradigms have produced models that achieve impressive state-of-the-art results on standard benchmarks, yet their relative representational capabilities across different levels of recognition granularity remain insufficiently explored. This research proposes a systematic comparative analysis between a leading model from each paradigm:

V-JEPA 2.1 (Assran et al., 2025), a self-supervised video encoder developed by Meta AI with 654 million parameters, and Qwen2.5-VL (Qwen Team, 2025), an instruction-tuned vision-language model with 7 billion parameters. The evaluation is conducted on two benchmark datasets: FineGym (fine-grained gymnastics event classification) and HMDB51 (coarse-grained human action recognition).

1.2 Problem Statement

Despite recent advancements in video foundation models, several critical research gaps persist in the literature. Existing studies evaluate self-supervised video encoders and vision-language models in isolation, using disparate datasets, evaluation protocols, and fine-tuning setups, and direct controlled comparisons of their representation quality under identical frozen-feature evaluation protocols are missing. Furthermore, while both paradigms excel on specific benchmarks, how their relative performance changes across different levels of task granularity—from fine-grained domain-specific actions (FineGym, 4 classes) to coarse-grained general actions (HMDB51, 51 classes)—has not been systematically investigated. V-JEPA 2.1 (654M parameters) is approximately $10\times$ smaller than Qwen2.5-VL (7B parameters), and it remains an open question whether massive parameter scale in general-purpose VLMs translates to superior video action representations compared to compact, video-specialized self-supervised encoders.

The problem is therefore not only whether frozen self-supervised latent prediction encoders can learn more discriminative temporal representations than large vision-language models, but whether compact, video-specialized architectures can match or exceed the representational quality of models with an order-of-magnitude more parameters across varying task granularities.

1.3 Objectives

The objectives of this study are:

1. To evaluate and compare the effectiveness of self-supervised video representations (V-JEPA 2.1) and vision-language representations (Qwen2.5-VL) for video action recognition across fine-grained (FineGym) and coarse-grained (HMDB51) task granularities.

2. To design, implement, and evaluate a specialized classification architecture HAF-STD Net (Hierarchical Adaptive Fusion with Spectral-Temporal Decomposition) tailored for frozen self-supervised video representations.

Chapter 2

Background Study and Literature Review

2.1 Background Study

Traditional self-supervised learning approaches in computer vision have relied either on generative pixel reconstruction or on contrastive learning across augmented views. Pixel reconstruction forces models to spend representational capacity on high-frequency, unessential visual details, while contrastive learning requires large batch sizes and carefully designed negative sampling. To address these limitations, Joint Embedding Predictive Architectures (JEPA) perform non-generative prediction directly within a learned latent feature space. Assran et al. (2023) introduced Image-JEPA (I-JEPA), demonstrating that predicting missing target representations from context patches in latent space produces highly semantically rich visual embeddings.

Extending the JEPA paradigm to temporal data, Bardes et al. (2024) proposed V-JEPA, which applies feature prediction to spatiotemporal video blocks. Subsequently, Assran et al. (2025) developed V-JEPA 2.1, scaling video latent prediction using a ViT-Huge backbone (654M parameters) trained on extensive unlabelled video collections. Most recently, Mur-Labadia et al. (2026) introduced V-JEPA 2.1, incorporating a dense predictive loss and multi-layer intermediate supervision to enhance spatial and temporal feature grounding. Variational extensions to JEPA have also been explored by Y. Huang (2026), who proposed Variational JEPA (VJEPA), introducing a probabilistic latent prediction framework that models distribution uncertainty for stochastic control and predictive world modeling.

Parallel to self-supervised feature learning, Vision-Language Models (VLMs) have emerged as powerful general-purpose foundation models. As surveyed by Bordes et al. (2024) and Y. Zhang et al. (2023), VLMs connect visual encoders with autoregressive language model decoders through visual instruction tuning. Recent developments in vision-language architectures include the Qwen2.5-VL technical report (Bai et al., 2025) and Qwen3-VL (Qwen Team, 2025), which support native spatiotemporal visual tokenization, variable frame resolution, and long-context multimodal understanding. To enable practical deployment under strict hardware resource boundaries, Pathak and Han (2026) reviewed quantization, pruning, and low-bit inference techniques for large

vision-language models.

2.2 Literature Review

The progression from Image-JEPA to V-JEPA 2.1 reveals a systematic trajectory of architectural refinement aimed at producing increasingly expressive latent representations. Assran et al. (2023) first established the foundational principle that predicting target patch representations from context patches—without pixel-level reconstruction—produces semantically richer embeddings than generative or contrastive alternatives. Their Image-JEPA demonstrated that a non-generative latent predictive objective avoids wasting model capacity on perceptually irrelevant high-frequency detail, while simultaneously eliminating the dependency on hand-crafted data augmentations inherent to contrastive learning pipelines.

Building on this image-level foundation, Bardes et al. (2024) extended the joint-embedding predictive paradigm to the spatiotemporal domain with V-JEPA, introducing masked spatiotemporal block prediction in video feature space. The critical insight was that temporal coherence across video frames provides an especially rich supervisory signal for latent prediction, as adjacent frames share semantic content while exhibiting continuous motion dynamics. Assran et al. (2025) then scaled the V-JEPA framework significantly with V-JEPA 2, employing a ViT-Huge backbone with 654 million parameters trained on large-scale unlabelled video data. This scaling demonstrated that increasing both model capacity and training data volume within the JEPA paradigm yields representations that transfer competitively across diverse downstream tasks.

The most recent architectural advancement, V-JEPA 2.1 (Mur-Labadia et al., 2026), introduced two critical innovations: a dense per-token prediction loss that supervises each masked spatiotemporal token individually, and multi-layer intermediate supervision applied across encoder depth. These modifications fundamentally change the nature of learned representations by ensuring that intermediate encoder layers also acquire strong semantic features, rather than concentrating representational quality only in the final layer. This property is especially relevant to the present study, as it motivates the extraction of multi-layer features for the proposed HAF-STD Net architecture. In parallel, Y. Huang (2026) proposed Variational JEPA, which reformulates latent prediction as a probabilistic inference problem by introducing distribution uncertainty modelling, demonstrating that JEPA-style architectures can be extended beyond deterministic prediction to support stochastic control and predictive world modeling.

Integrating JEPA principles into multimodal and language domains represents an active frontier of research. Chen et al. (2026) proposed VL-JEPA, extending joint-embedding predictive learning to multimodal vision-language alignment without relying on autoregressive token generation. By performing cross-modal latent predic-

tion between visual and textual representations, VL-JEPA demonstrated that the non-generative JEPA objective can bridge modality gaps while preserving the computational advantages of avoiding pixel or token reconstruction. H. Huang et al. (2025) combined Large Language Models with JEPA, demonstrating latent-space predictive loss objectives for language understanding. Their LLM-JEPA framework showed that large language models benefit from predictive representation learning in ways that complement standard autoregressive training. Liu et al. (2026) presented JEPA-Reasoner, decoupling latent-space reasoning from discrete token generation to enable unconstrained reasoning paths, while H. Zhang et al. (2026) introduced ThinkJEPA, empowering latent world models with large vision-language reasoning capabilities. Collectively, these works indicate a growing trend toward unifying self-supervised prediction with language grounding, though none directly evaluate how these latent representations compare against instruction-tuned VLMs for discriminative video understanding tasks.

Beyond visual recognition, JEPA architectures serve as foundational world models for physical reasoning and decision-making. Bagatella et al. (2025) developed Temporal Difference JEPA, proving that latent-predictive representations enable zero-shot reinforcement learning across complex control environments. By combining temporal difference learning with JEPA's latent prediction, TD-JEPA demonstrated that forward predictive representations in a learned feature space encode sufficient dynamical information for effective policy transfer without environment-specific fine-tuning. Terver and Yang (2026) investigated the core drivers of success in physical planning using joint-embedding predictive world models, demonstrating strong generalization in physical simulation tasks. Their analysis identified that the latent prediction objective, rather than model scale alone, is the primary contributor to planning performance, providing evidence that JEPA representations capture causal-temporal structure that is qualitatively different from the associative representations learned through vision-language pretraining.

A fundamental distinction exists between the representation learning strategies of JEPA-based self-supervised models and instruction-tuned vision-language models. Self-supervised JEPA encoders such as V-JEPA 2.1 (Mur-Labadia et al., 2026) learn visual representations through prediction in latent space, entirely without textual supervision. Their learned features are inherently visual-geometric, capturing motion trajectories, spatial configurations, and temporal dynamics as distributed patterns within a continuous embedding space. In contrast, vision-language models such as Qwen2.5-VL (Bai et al., 2025; Qwen Team, 2025) acquire visual representations through cross-modal alignment with natural language, producing features that are linguistically grounded and semantically structured according to textual category boundaries. As discussed in the comprehensive survey by Bordes et al. (2024), this linguistic grounding provides VLMs with strong zero-shot generalization capabilities, since visual concepts are inherently

linked to compositional language descriptions. Recent efforts have sought to bridge the gap between linguistic and geometric representations; notably, Lin et al. (2026) introduced Qwen-3D, a generalist 3D vision-language model built upon the Qwen backbone that augments visual tokens with 3D Rotary Positional Embeddings and operates attention directly in 3D scene space, enabling scalable cross-view and temporal reasoning through explicit geometric awareness rather than purely linguistic alignment. However, Y. Zhang et al. (2023) noted that vision-language alignment may bias representations toward semantically salient concepts that are easily described in natural language, potentially sacrificing fine-grained visual distinctions that lack clear linguistic labels.

The efficiency dimension further differentiates these paradigms. V-JEPA 2.1, with 654M parameters, achieves its representational power through architectural specialization and self-supervised pretraining on video data alone, whereas Qwen2.5-VL requires 7B parameters and massive multimodal corpora to achieve general-purpose visual understanding. Pathak and Han (2026) surveyed quantization and pruning strategies for deploying large VLMs under hardware constraints, highlighting that the parameter efficiency of specialized self-supervised encoders represents a significant practical advantage for resource-constrained deployment scenarios.

While self-supervised JEPA models (V-JEPA 2.1, (Assran et al., 2025); V-JEPA 2.1, (Mur-Labadia et al., 2026)) and large vision-language models (Qwen2.5-VL, (Qwen Team, 2025)) have demonstrated impressive individual capabilities, previous literature lacks a direct, controlled empirical comparison of their frozen-feature representations across fine-grained and coarse-grained video action recognition tasks. Furthermore, existing probing protocols rely on simple global average pooling, ignoring the rich multi-layer intermediate depth representations available in JEPA backbones. This study directly addresses these gaps by evaluating both representation paradigms across 11 baseline classifiers and introducing HAF-STD Net architecture to exploit multi-layer temporal dynamics.

Chapter 3

Materials and Methods

3.1 Overview of Research Methodology

This research adopts a rigorous, controlled frozen-feature evaluation protocol to compare the representational capabilities of self-supervised video encoders (V-JEPA 2.1) against instruction-tuned vision-language models (Qwen2.5-VL). In addition to baseline evaluation, we use a specialized downstream classification architecture HAF-STD Net (Hierarchical Adaptive Fusion with Spectral-Temporal Decomposition) designed to explicitly exploit V-JEPA 2.1's temporal token coherence and hierarchical depth representation space.

The methodology follows a multi-stage experimental pipeline: (1) Data Acquisition and Sampling, (2) Spatiotemporal Multi-Layer Feature Extraction, (3) Feature Preprocessing and Dimensionality Reduction, (4) Downstream Evaluation across 11 Baseline Classifiers and HAF-STD Net, and (5) Comparative Analysis.

Figure 3.1 provides a high-level flowchart depicting the end-to-end experimental architecture.

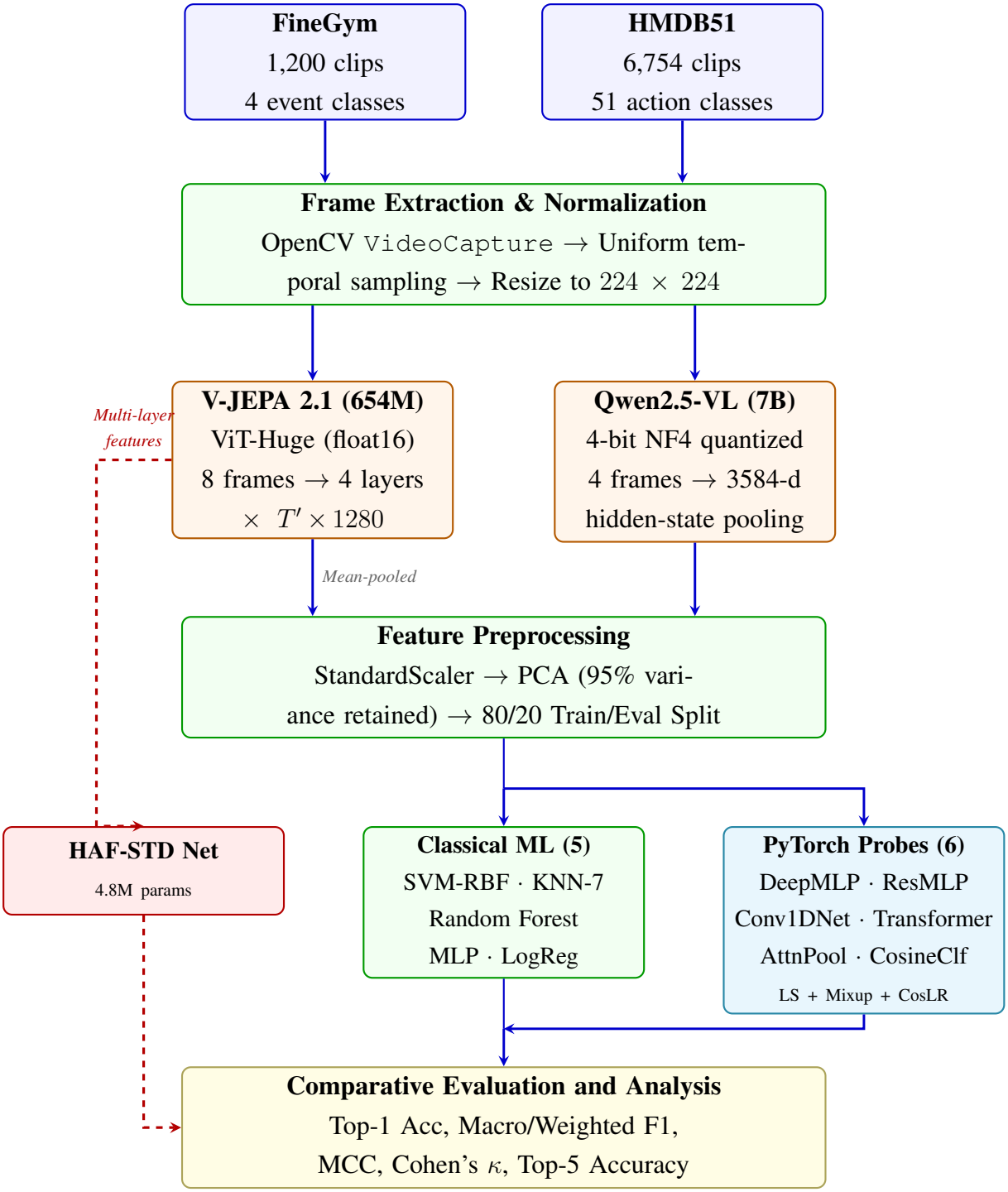


Figure 3.1: Flowchart depicting the proposed multi-stage experimental methodology pipeline. Solid arrows indicate the shared baseline path where mean-pooled features from both models undergo PCA preprocessing before evaluation by 11 classifiers. Dashed red arrows indicate the HAF-STD Net path, which operates exclusively on V-JEPA 2's multi-layer temporal features without PCA reduction.

3.2 Data Acquisition and Dataset Characteristics

To evaluate model performance across different task granularities, two benchmark datasets of contrasting complexity are utilized:

3.2.1 FineGym Dataset (Fine-Grained Action Recognition)

FineGym is a fine-grained activity recognition benchmark containing high-frame-rate competitive gymnastics footage. The official annotation file (v1.1) specifies precise temporal boundaries and hierarchical activity labels. In this study, evaluation focuses on the top-level Event classification tier, comprising 4 distinct gymnastics apparatus events: Balance Beam (BB), Floor Exercise (FX), Uneven Bars (UB), and Vault (VT). A balanced sample of 300 clips per class (1,200 total clips) is targeted for download via `yt-dlp` from YouTube.

3.2.2 HMDB51 Dataset (General Action Recognition)

HMDB51 is a classic benchmark for general human action recognition, curated from commercial movies and web videos. It comprises 6,754 video clips categorized into 51 action classes, encompassing facial expressions, general body motion, human-object interactions, and human-human interactions. The complete dataset is ingested directly from the Hugging Face Hub repository (`jili5044/hmdb51`).

Table 3.1 summarizes the structural properties and comparative parameters of both benchmark datasets.

Table 3.1: Comparison of FineGym and HMDB51 benchmark dataset specifications.

Dataset Parameter	FineGym (Event Level)	HMDB51
Task Granularity	Fine-grained (Gymnastics)	Coarse-grained (General)
Number of Classes	4	51
Target Sample Size	1,200 clips	6,754 clips
Domain Context	Uniform arena/gym environment	Diverse real-world settings
Primary Discriminative Cue	Subtle body kinematics	Broad motion & background
Evaluation Split	80% Train / 20% Test	80% Train / 20% Test

3.3 Data Preprocessing and Input Pipeline

Video clips undergo standardized preprocessing prior to feature extraction:

1. **Uniform Temporal Frame Sampling:** For a video clip containing N total frames, a sequence of N_f frames is uniformly sampled along the temporal axis:

$$t_i = \left\lfloor i \cdot \frac{N}{N_f} \right\rfloor, \quad i \in \{0, 1, \ldots, N_f - 1\}$$

   For V-JEPA 2.1, $N_f = 8$ frames are sampled; for Qwen2.5-VL, $N_f = 4$ frames are sampled.

2. **Spatial Resizing and Normalization:** Sampled frames are resized to 224×224 pixels and normalized using standard ImageNet mean ($\boldsymbol{\mu} = [0.485, 0.456, 0.406]$) and standard deviation ($\boldsymbol{\sigma} = [0.229, 0.224, 0.225]$).

3.4 Feature Extraction Framework

Both candidate models function strictly as frozen feature extractors without updating backbone parameters.

3.4.1 Self-Supervised Encoder: V-JEPA 2.1

V-JEPA 2.1 (Assran et al., 2025) utilizes a ViT-Huge backbone (654M parameters) trained via non-generative spatiotemporal prediction in a latent embedding space. Given an input video sequence of 8 sampled frames, we extract two feature representations:

1. **Multi-Layer Per-Frame Features (for HAF-STD Net):** Hidden states are extracted from 4 intermediate encoder layers at 25%, 50%, 75%, and 100% depth. Spatial tokens are mean-pooled per frame, producing a tensor $\mathbf{H}_{\text{multi}} \in \mathbb{R}^{4 \times T' \times 1280}$ per clip that preserves both depth hierarchy and temporal structure.

2. **Global Mean-Pooled Features (for Baseline Probes):** Global average pooling across all spatiotemporal tokens of the final layer outputs $\mathbf{f}_{\text{VJEPA}} \in \mathbb{R}^{1280}$.

3.4.2 V-JEPA 2.1 Self-Supervised Training Objectives

The representational quality of the frozen V-JEPA 2.1 encoder stems from its self-supervised pretraining objective, which performs *mask-denoising prediction in representation space* rather than pixel reconstruction. Given two views x and y of a video, where patches have been randomly dropped from y, the encoder $E_\theta(\cdot)$ extracts representations from the unmasked view x, and a predictor $P_\phi(\cdot)$ predicts the representation of the masked view. The training objective minimizes the ℓ_1 distance between predicted and target representations (Assran et al., 2025):

$$\underset{\theta, \phi, \Delta_y}{\text{minimize}} \quad \| P_\phi(\Delta_y, E_\theta(x)) - \text{sg}(E_{\bar{\theta}}(y)) \|_1$$

where Δ_y is a learnable mask token that indicates the locations of the dropped patches, $\mathrm{sg}(\cdot)$ denotes a stop-gradient operator to prevent representation collapse, and $\bar{\theta}$ represents an exponential moving average (EMA) of the encoder weights θ. The loss is applied only to predictions of masked patches, and both the encoder $E_\theta(\cdot)$ and predictor $P_\phi(\cdot)$ are parameterized as Vision Transformers (ViTs) with Rotary Position Embedding (RoPE).

V-JEPA 2.1 (Mur-Labadia et al., 2026) refines this objective by introducing a *dense per-token prediction loss* that applies supervision individually to each masked spatiotemporal token rather than globally:

$$\mathcal{L}_{\text{predict}} = \frac{1}{|M|} \sum_{i \in M} \| P_\phi(E_\theta(x), \Delta_y)_i - \mathrm{sg}(E_{\bar{\theta}}(y)_i) \|_1$$

where M is the set of masked patch indices from the x view, and the subscript i denotes the i-th token position. This dense formulation, combined with multi-layer intermediate supervision across encoder depth, enhances the spatial and temporal grounding of learned representations.

Figure 3.2 illustrates the complete V-JEPA 2.1 pretraining architecture. The dual-encoder framework consists of an x-Encoder E_θ that processes unmasked context tokens and a y-Encoder $E_{\bar{\theta}}$ (updated via EMA) that produces target representations. A multi-level predictor P_ϕ reconstructs the masked spatiotemporal token representations by fusing features across multiple encoder depths, with two complementary loss terms: a context loss ($\lambda_d \cdot L_1$) and a prediction loss (L_1), both applied through stop-gradient operations to prevent representational collapse.

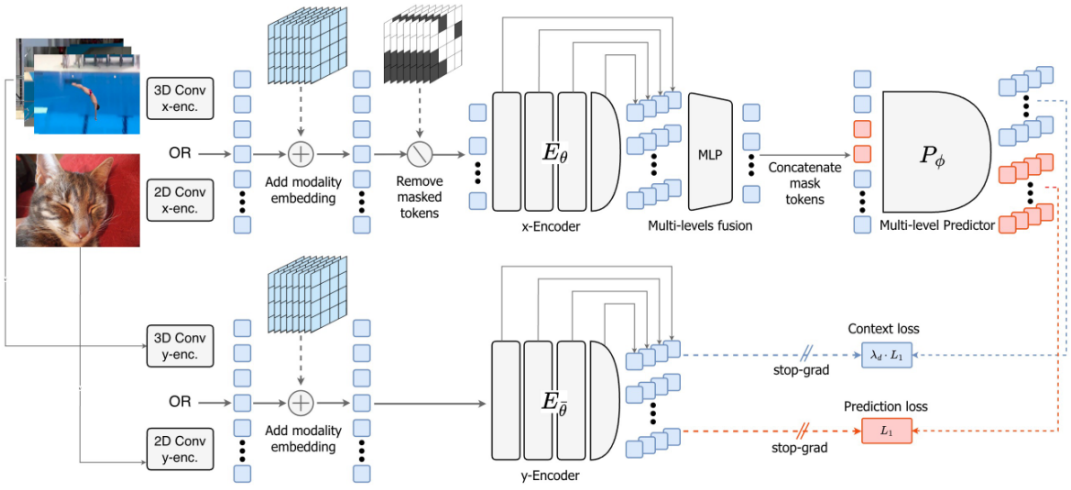


Figure 3.2: Architecture of V-JEPA 2.1 self-supervised pretraining framework showing the dual x-Encoder and y-Encoder pipeline with multi-level fusion, mask-denoising prediction, and stop-gradient targets (Mur-Labadia et al., 2026).

3.4.3 Vision-Language Model: Qwen2.5-VL

Qwen2.5-VL (Qwen Team, 2025) (7B parameters) integrates a Vision Transformer encoder with a language model decoder. The model is loaded in 4-bit NormalFloat (NF4) quantization (Pathak & Han, 2026). The 4 sampled frames are passed alongside a minimal prompt (`"Describe the video action."`). The output hidden state of the final decoder layer $\mathbf{H}_{\text{decoder}} \in \mathbb{R}^{S \times 3584}$ is average-pooled across sequence tokens:

$$\mathbf{f}_{\text{Qwen}} = \frac{1}{S} \sum_{s=1}^{S} \mathbf{h}_s \in \mathbb{R}^{3584}$$

Figure 3.3 depicts the Qwen2.5-VL architecture. The model employs a Vision Encoder with a Conv3D patch embedding layer and window-partitioned attention blocks, which processes images and video frames at their native resolution. Visual tokens are then projected and interleaved with text tokens into the Qwen2.5 Language Model (LM) Decoder for autoregressive generation. Notably, the architecture uses Multimodal Rotary Position Embedding (MRoPE) with absolute time alignment for temporal grounding of video inputs and dynamic FPS sampling to accommodate variable-length video sequences.

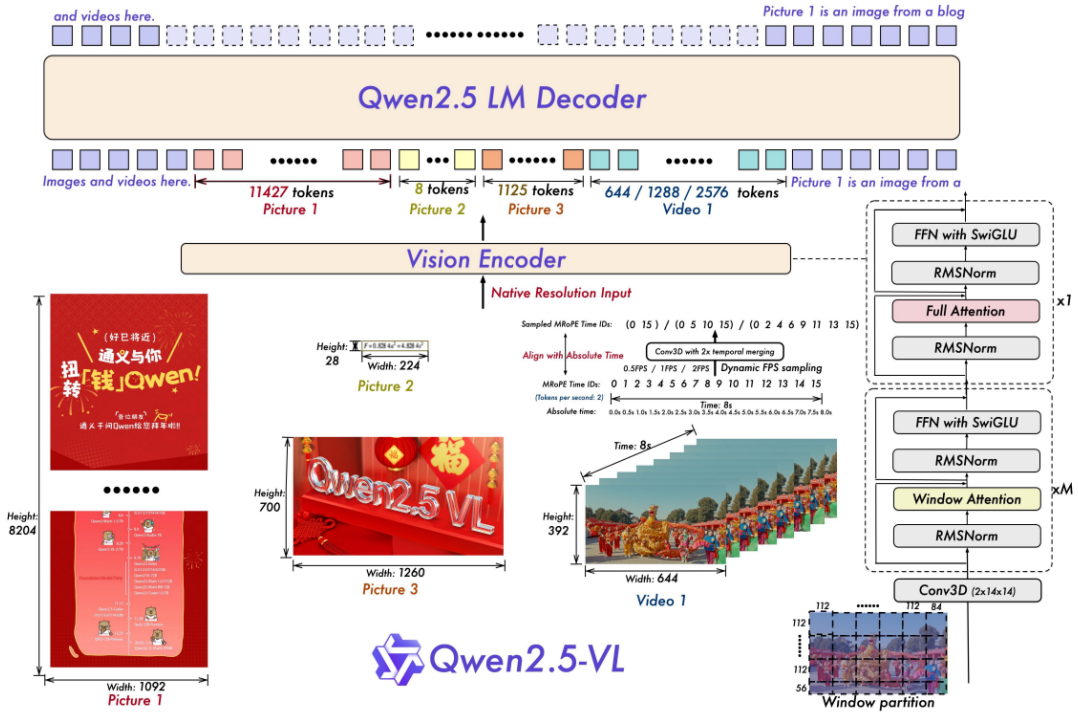


Figure 3.3: Architecture of Qwen2.5-VL vision-language model illustrating the Vision Encoder with Conv3D patch embedding, window attention, and native resolution input processing, feeding into the Qwen2.5 LM Decoder with MRoPE temporal alignment (Qwen Team, 2025).

3.5 Feature Standardization and Dimensionality Reduction

To ensure optimal classifier convergence and mitigate the curse of dimensionality, extracted feature vectors undergo standard transformations:

1. **Z-Score Standardization:** Feature vectors $\mathbf{x}_i$ are standardized using training set statistics:
$$\mathbf{z}_i = \frac{\mathbf{x}_i - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}}}$$

2. **Principal Component Analysis (PCA):** PCA is fitted on the standardized training feature matrix $\mathbf{Z}_{\text{train}}$ to project representations onto an orthogonal subspace while retaining 95% of total variance:
$$K = \arg\min_k \left\{ \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{m=1}^{D} \lambda_m} \geq 0.95 \right\}$$

3.6 Downstream Baseline Classifiers

An ensemble of 11 baseline classifiers (5 classical ML and 6 PyTorch neural network probes) is evaluated on the transformed feature representations:

3.6.1 Classical Machine Learning Classifiers

1. **Support Vector Machine (SVM-RBF):** RBF kernel with $C = 10.0$ and balanced class weighting.

2. **k-Nearest Neighbors (k-NN):** Euclidean distance matching with $k = 7$ neighbors.

3. **Random Forest:** Ensemble of 500 decision trees with Gini impurity split criteria.

4. **Multi-Layer Perceptron (MLP):** Scikit-learn MLP with two hidden layers (512, 256) and adaptive learning rate.

5. **Softmax Logistic Regression:** Multinomial linear probe with L2 regularization ($C = 1.0$).

3.6.2 PyTorch Neural Network Probe Architectures

1. **DeepMLP:** 4-layer feedforward network (512 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ Classes) with BatchNorm, GELU, and Dropout (0.3).

2. **ResMLP:** Deep residual MLP with 3 residual blocks: $\mathbf{h}_{l+1} = \text{GELU}(\mathbf{h}_l + \mathcal{F}(\mathbf{h}_l))$.

3. **Conv1DNet:** 1D CNN treating feature dimensions as sequence channels.

4. **MiniTransformer:** 2-layer Transformer Encoder with multi-head self-attention ($h = 4$) on chunked feature embeddings.

5. **AttentionPoolMLP:** Learnable context query attention pooling mechanism.

6. **CosineClassifier:** Magnitude-invariant prototype classifier computing scaled cosine similarity with temperature $\tau = 16.0$.

All PyTorch baseline models are trained using label smoothing ($\epsilon = 0.1$), feature-space mixup ($\alpha = 0.2$), Cosine Annealing LR scheduling, and early stopping.

3.7 Proposed Classifier Architecture: HAF-STD Net

As a key technical contribution of this thesis, we propose HAF-STD Net (Hierarchical Adaptive Fusion with Spectral-Temporal Decomposition), a classification architecture (4.8M parameters) specifically tailored for frozen V-JEPA 2.1 backbones. HAF-STD Net operates on multi-layer per-frame features $\mathbf{H} \in \mathbb{R}^{4\times T'\times 1280}$ and incorporates three core mechanisms:

3.7.1 1. Spectral-Temporal Decomposition via 1D Haar DWT

At each selected layer $k \in \{1, 2, 3, 4\}$, a 1D Haar Discrete Wavelet Transform is applied along the temporal dimension T':

$$\mathbf{a}_k = \frac{1}{2}(\mathbf{h}_{2t} + \mathbf{h}_{2t+1}), \quad \mathbf{d}_k = \frac{1}{2}(\mathbf{h}_{2t} - \mathbf{h}_{2t+1})$$

where $\mathbf{a}_k$ isolates low-frequency static appearance features and $\mathbf{d}_k$ isolates high-frequency motion dynamics.

3.7.2 2. Inter-Layer Representation Velocity

Inter-layer deltas are computed between consecutive encoder layers to quantify representation velocity across depth:

$$\Delta\mathbf{H}_k = \bar{\mathbf{H}}_{k+1} - \bar{\mathbf{H}}_k$$

where $\bar{\mathbf{H}}_k$ is the temporally mean-pooled representation at layer k.

3.7.3 3. Adaptive Routing and Cosine Classification

The 3 streams per layer (appearance, motion, velocity) yield $4 \times 3 = 12$ projected feature streams $\mathbf{s}_i \in \mathbb{R}^{256}$. An adaptive routing network computes input-dependent

softmax weights $\mathbf{w} \in \mathbb{R}^{12}$:

$$\mathbf{w} = \text{softmax}\left(\text{MLP}\left([\mathbf{s}_1; \mathbf{s}_2; \ldots; \mathbf{s}_{12}]\right)\right), \quad \mathbf{f} = \sum_{i=1}^{12} w_i \cdot \mathbf{s}_i$$

The dynamically fused feature $\mathbf{f}$ is classified via a cosine-similarity head with temperature scaling τ:

$$P(y = c \mid \mathbf{f}) = \frac{\exp\left(\tau \cdot \hat{\mathbf{f}}^T \hat{\mathbf{w}}_c\right)}{\sum_{j=1}^{C} \exp\left(\tau \cdot \hat{\mathbf{f}}^T \hat{\mathbf{w}}_j\right)}$$

3.8 Performance Evaluation Metrics

Model performance across both datasets is evaluated using standard metrics:

- **Macro F1-Score:** Unweighted mean of class F1-scores.
- **Matthews Correlation Coefficient (MCC):** Balanced metric robust to class imbalance.
- **Cohen's Kappa (κ):** Chance-corrected classification agreement.

Experiments are conducted on Kaggle cloud platform using NVIDIA Tesla T4 GPUs (16 GB VRAM) with PyTorch 2.1, Hugging Face Transformers, and Scikit-Learn.

Chapter 4

Work Plan, Expected Outcomes, and Budgeting

4.1 Work Plan of the Study

The proposed research will be conducted over a 14-week timeline. Figure 4.1 illustrates the Gantt chart detailing the proposed project phases and execution timeline.

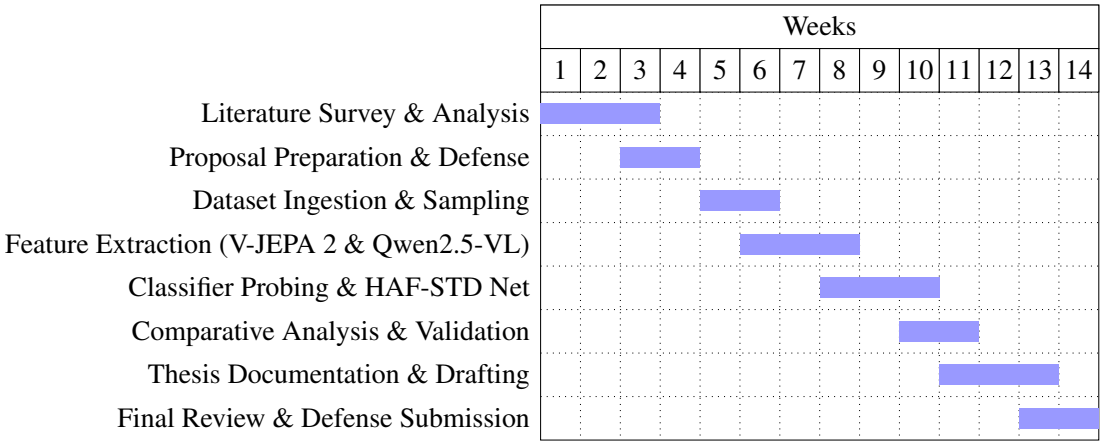


Figure 4.1: Gantt Chart illustrating the 14-week proposed research timeline.

Table 4.1 details the core activities and milestones planned for each phase of the research.

Table 4.1: Detailed timeline breakdown and major project milestones.

Phase	Timeline	Description of Activities
Literature Survey	Weeks 1–3	Comprehensive literature review of SSL video encoders, VLMs, fine-grained activity benchmarks, and gap analysis.
Proposal Defense	Weeks 3–4	Preparation, submission, and defense of the Master's thesis proposal document.
Data Acquisition	Weeks 5–6	Automated ingestion of FineGym clips via `yt-dlp` and extraction of HMDB51 dataset from Hugging Face Hub.
Feature Extraction	Weeks 6–8	Batch extraction of multi-layer and global features from V-JEPA 2.1 and Qwen2.5-VL on Kaggle Tesla T4 GPU.
Classifier	Weeks 8–10	Feature standardization, PCA fitting, training 11 baseline classifiers (5 classical ML + 6 PyTorch probes) and developing the **HAF-STD Net** architecture.
Statistical Analysis	Weeks 10–11	Cross-granularity performance comparison, HAF-STD Net vs baseline probe ablation, error matrix evaluation.
Documentation	Weeks 11–13	Comprehensive drafting of Master's thesis chapters, figures, tables, and appendices.
Thesis Submission	Weeks 13–14	Plagiarism verification, peer review, supervisor approval, and final thesis defense submission.

4.2 Expected Outcomes

The primary expected outcomes of this research project include:

1. **Empirical Feature Quality Evaluation:** Quantitative performance benchmarks comparing frozen representations from self-supervised video encoders (V-JEPA 2.1, 654M) against vision-language models (Qwen2.5-VL, 7B).

2. **Proposed Classifier Contribution (HAF-STD Net):** A classification architecture (4.8M parameters) combining 1D Haar wavelet spectral-temporal decomposition, depth representation velocity, and adaptive routing for frozen self-supervised backbones.

3. **Granularity-Dependent Representation Insights:** Detailed comparative evidence demonstrating how self-supervised latent prediction vs. multimodal language pre-training performs across fine-grained (FineGym) vs. coarse-grained (HMDB51) action classification.

4. **Parameter Efficiency Analysis:** Practical guidelines establishing whether compact, video-specialized encoders can outperform general-purpose 7B vision-language models on specialized motion tasks.

5. **Classifier Compatibility Assessment:** Identification of optimal classical machine learning models and PyTorch neural probes (ResMLP, DeepMLP, HAF-STD Net) for downstream representation spaces.

6. **Publication Artifacts:** A complete Master's thesis report, open-source experimental codebase, and clean dataset feature benchmarks for future research.

4.3 Budgeting and Computational Resources

Table 4.2 outlines the estimated hardware resource allocation and computational budget required for executing the proposed research work.

Table 4.2: Estimated computational resource allocation and budget.

Resource Item	**Specification / Provider**	**Cost (NPR)**
GPU Compute Acceleration	Kaggle Cloud Environments (NVIDIA Tesla T4 16GB VRAM)	Free Tier
Storage & Dataset Bandwidth	Cloud Storage / High-Speed Academic Network (TU)	Academic Access
Software & Libraries	Open-source PyTorch, Hugging Face, Scikit-Learn	N/A (Open Source)
Documentation & Printing	Proposal & Thesis Binding (As per IoST Regulations)	NRs. 2,500
Total Estimated Budget		**NRs. 2,500**

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